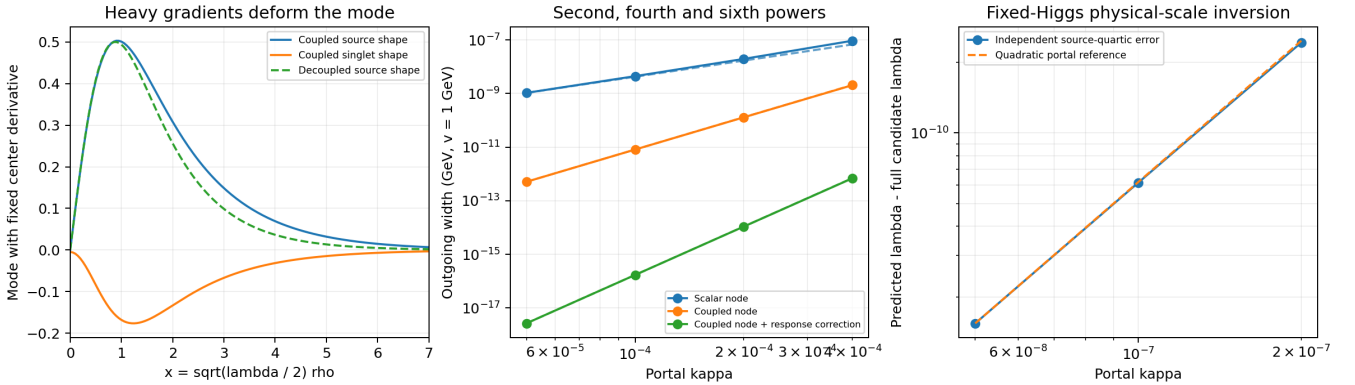


Finite Gaussian mediation and the coupled radiation-node hierarchy

The polynomial wall-response construction now includes the finite Gaussian mediator. The mediator deforms the source wall and its localized shape before the Higgs portal is switched on, so the decoupled scalar radiation node is generally displaced. The new calculation derives that coupled node, its first portal correction and its nonzero tuning multiplier without fitting a nonlinear localized candidate or an outgoing width. It checks the heavy-mode profile against an independent Green convolution, the tuning multiplier against a Fourier derivative identity, and the portal correction against full three-field solves. The resulting leakage sequence is second, fourth and sixth power in the portal for the scalar node, corrected coupled leading node and coupled node with its first response correction, respectively. The physical-scale fixed-Higgs inversion predicts the earlier 154 GeV candidate's source quartic within a few parts per million at the recorded portal.



The full action is retained. In $\rho = v z$, define $\alpha = g/v$, $\mu = M/v$ and $h_0 = v_H/v$. The source field u is odd; the singlet y and Higgs radius h are even. The dimensionless potential is

$$V = \frac{\lambda}{4}(u^2 - 1)^2 + \frac{\mu^2}{2}\left(y + \frac{\alpha u^2}{\mu^2}\right)^2 + \frac{\lambda_H}{4}\left[h^2 - h_0^2 + \frac{\kappa}{\lambda_H}(u^2 - 1)\right]^2.$$

At zero portal, solve the source and singlet wall with $h = h_0$. This calculation retains the singlet gradient energy and its boundary equation; it does not substitute $y = -\alpha u^2 / \mu^2$ pointwise. Let $r_s = y + \alpha u^2 / \mu^2$. The two source-sector Hessian entries are $K_{uu} = \lambda(3u^2 - 1) + 2\alpha r_s + 4\alpha^2 u^2 / \mu^2$, $K_{uy} = 2\alpha u$ and $K_{yy} = \mu^2$. The localized shape has odd source component P and even singlet component S , with source derivative one at the center. Its energy E_0 must lie below both coupled source vacuum thresholds. The current continuation searches the shape connected to the scalar reference and a leading radiation node near the scalar zero; it is not a census of all bound modes or all radiation zeros.

Exact nonlocal heavy elimination. The heavy shape equation is $(-\partial_\rho^2 + \mu^2 - E_0)S = -2\alpha uP$. Since E_0 is below the heavy threshold, its inverse is a positive decaying Green operator. Eliminating S gives the source equation

$$\left[-\partial_\rho^2 + K_{uu} - E_0 - 4\alpha^2 u(-\partial_\rho^2 + \mu^2 - E_0)^{-1}u\right]P = 0.$$

Both occurrences of u are multiplication operators; the inverse between them must not be commuted through them. The source spectrum consequently depends on a nonlocal, energy-dependent response. In one dimension the heavy kernel is $\exp(-\sqrt{\mu^2 - E_0}|\rho - t|)/(2\sqrt{\mu^2 - E_0})$. Even parity adds the reflected kernel on the positive half-line. The independent numerical convolution uses this kernel rather than the coupled boundary solver. Its exponentially narrow peak is integrated after scaling the distance by its decay constant, so the physical large-heavy-mass case remains resolved. In the stronger coupled example the singlet mode at the center is about -0.0047188 in the dimensionless shape normalization, while a local valley replacement would give zero because u and P both vanish there. Across five sample points the coupled mode and explicit convolution differ by less than two times ten to the minus fourteen.

The radiation-node problem. Put $k=\sqrt{\lambda/2}$, $x=k\rho$, $p=kP$ and $e_0=E_0/k^2$. Let q be the open Higgs wave number divided by k . At leading portal order the induced Higgs response is $\chi=k h_0 G(x)/k^3$. It satisfies $G\text{-double-prime}+q^2 G=2u p$. A real even solution decaying at both ends exists at this order when

$$F(q) = \int_{-\infty}^{\infty} \cos(qx) u(x) p(x) dx = 0, \quad \frac{m_{H,0}^2}{v^2} = k^2(e_0 - q^2).$$

This is the coupled replacement for the scalar polynomial form-factor zero. In the scalar limit $u=\tanh(x)$, $p=\text{sech}(x)\tanh(x)$, $q=1$ and $e_0=3$, so the unmixed Higgs threshold is λ . With finite mediation the source profiles and energy are different, and the original polynomial radiation zero is no longer the correct node. The root search records a real cosine sign bracket inside the open Higgs interval. A separate forced boundary solve determines q and the localized response G , with parity at zero and zero value and derivative at the finite endpoint. These two determinations of q agree within the recorded numerical controls. Their agreement is not an infinite-domain existence certificate.

The Higgs background and the next cancellation. Define $m=e_0-q^2$. The first Higgs background response is $h=h_0+k h_0 H/k^2+O(k^2)$, where $H\text{-double-prime}=mH(1-u^2)$. The source and singlet backgrounds have no first-order portal forcing, because the square-completed portal couples to $h^2-h_0^2$, which vanishes at zero portal. The source shape and energy likewise have no first-order portal correction. These facts are checked by exact symbolic differentiation of the complete three-field potential, including its Gaussian square. Write the adjusted unmixed threshold as $m_H^2/v^2=k^2 m+C_G k^2+O(k^2)$. The second Higgs-mode response B obeys

$$B'' + q^2 B = 2uHp + (3mH + u^2 - 1 + C_G)G.$$

Its real cosine solvability condition selects

$$C_G = -\frac{\int \cos(qx) [2uHp + (3mH + u^2 - 1)G] dx}{\int \cos(qx) G dx}.$$

All common powers of h_0 and k cancel. Thus C_G remains independent of h_0 when the zero-portal source action is held fixed, even though it depends on the Gaussian source profiles and spectrum. The calculation obtains C_G both as the free parameter of this forced boundary problem and from the independent overlap ratio. Neither determination uses the full nonlinear candidate curve or outgoing widths as fitted input.

A transversality identity. Fourier transforming the leading response gives $(q^2-xi^2) \hat{G}(xi)=2F(xi)$. At the radiation zero, both numerator and denominator vanish. Taking the limit at $xi=q$ yields

$$\int \cos(qx) G(x) dx = -\frac{F'(q)}{q}.$$

This connects the tuning multiplier to the derivative of the leading radiation form factor. It is also a practical independent check: one side integrates the localized forced mode, while the other differentiates the source overlap using $-x \sin(qx)u p$. A nonzero derivative makes the leading radiation zero simple and permits one real threshold correction to cancel the next real radiation amplitude. In the strong Gaussian example both sides are about 1.68427141251. The multiplier is therefore numerically well separated from zero in that case. This is evidence for the local continuation and a formal solvability relation, not an assertion that every possible Gaussian action has a nondegenerate node.

The quantitative coupled example. Use $v=1$ GeV, source $\lambda=0.1$, heavy $\mu=3$ and $\alpha=2$. The zero-portal shape energy is approximately 0.1367406806594, divided by v^2 . The coupled leading unmixed Higgs threshold is approximately 0.0970420278296, rather than the scalar value 0.1. The first portal coefficient is approximately -1.44030786942, rather than -1.36468352108. A weaker $\alpha=0.1$, $\mu=10$ case gives leading threshold about 0.0999999462549 and coefficient about -1.36468496099. The zero-Gaussian calculation recovers the scalar threshold, shape energy and universal slope. A coupling scan and domain/tolerance controls retain these values and residuals separately.

Independent nonlinear continuation. Four joint wall/mode solves at portals 0.00005, 0.0001, 0.0002 and 0.0004 provide the quotient $[\text{full threshold-coupled zero-portal threshold}]/k^2$. It approaches the derived C_G as the portal decreases.

At portal 0.0001 the quotient is about -1.43371216; its difference from the predicted -1.44030787 is a higher-order correction, not a fit used to construct the response. The source background, singlet background and full three-field Hessian in these controls all retain the finite Gaussian coupling.

What the mediator changes about leakage. If the scalar threshold λ is reused, the source overlap at the actual open wave number is nonzero. The first radiation amplitude is then order κ and the width order κ^2 . Moving to the coupled leading threshold cancels that amplitude, leaving fourth-power leakage. Adding $C_G \kappa$ cancels the next amplitude and leaves sixth-power leakage. In the stronger coupled example, successive portal doubling ratios approach 4, 16 and 64 in these three prescriptions. At portal 0.0001 the corresponding independent outgoing widths are approximately 4.44962 times ten to the minus nine, 8.14285 times ten to the minus twelve and 1.71069 times ten to the minus sixteen GeV. The smallest-portal doubling ratios in the full receipt are about 4.202, 15.967 and 63.885. This separates a mediator-induced leading radiation channel from higher-order leakage restored by Higgs background distortion. The weak Gaussian limit has a small leading amplitude, but the smallness of its coefficient does not change its asymptotic portal order.

The fixed-Higgs source inversion. Hold $v=30000$ GeV, $M=300000$ GeV, $g=3000$ GeV, $v_H=246$ GeV and $\lambda_H=0.13$ fixed. The target unmixed Higgs threshold is $2\lambda_H h_0^2$. Instead of varying the Higgs quartic, solve the derived leading-plus-linear node equation for the source quartic. At portal ten to the minus seven, this predicts source λ approximately 1.76188778979 times ten to the minus five. The independent full localized-candidate solve selects approximately 1.76188164948 times ten to the minus five and mode mass approximately 154.2250242 GeV. Their source-quartic difference is approximately 6.14031 times ten to the minus eleven, or about 3.5 parts per million. At half and twice the portal the errors are approximately 1.54441 times ten to the minus eleven and 2.42786 times ten to the minus ten; the near-fourfold ratios are consistent with the omitted quadratic portal term. The full candidate values are comparison outputs, not inputs to the source inversion.

Algebra, execution and numerical meaning. Seven exact rational-symbolic identities check the first source and heavy portal zeros, the Higgs background forcing, the vanishing direct singlet-Higgs Hessian, the two open-mode forcings and the Fourier transversality limit. Ten focused tests check those identities, the scalar limit, source/continuum thresholds, independent moments, heavy convolution, finite-gradient effects, domain and Higgs-ratio controls, nonlinear continuation, sixth-power outgoing leakage and fixed-Higgs prediction. Sixty-three related wall tests also pass. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_gaussian_fredholm.py` for the scientific receipt, `python scripts/plot_wall_gaussian_fredholm.py` for the figure and `python scripts/render_wall_gaussian_fredholm.py` for this illustrated account. The new code occupies separate wall files and preserves the other sessions' Weil, arithmetic and flavor results.

The response domain controls use $x=24,28,32$ with tighter tolerances. In the strong Gaussian example, the last two leading thresholds differ by about five times ten to the minus eleven, and the first portal coefficients by about two times ten to the minus eight. The induced open responses use zero endpoint value and derivative; the heavy/source shape uses vacuum-channel decaying conditions. Closed components and source forcing still have finite tails, so boundary residuals and numerical quadrature are not certified enclosures. The independent heavy convolution uses an exponential-distance cutoff of 50 and reports its numerical comparison, without claiming a rigorous tail error. The selected coupled node and its correction do not prove infinite-domain existence, global vacuum selection or quantum persistence.

The mathematical extension is the move from the scalar polynomial factor to a finite-mediator, nonlocal shape operator whose simple Fourier zero controls radiation cancellation. The previously derived Gaussian-free quadratic portal coefficient is not silently assigned to this coupled action; that next coefficient remains uncomputed here. A full coupled higher-order hierarchy needs the source/singlet static corrections, their normalized shape response and the corresponding energy moment. Convergence estimates, actual infinite-domain radiation zeros, gauge and fermion leakage, and a justified flavor symmetry enforcing the tuned relation remain subsequent work. The 154 GeV object continues to be a finite-radius localized candidate, and changing its source quartic prevents transferring the earlier benchmark's thermal and network conclusions unchanged.